

CREATE A CHATBOT IN PYTHON

INTRODUCTION TO CHATBOT:

A chatbot also known as a chatterbot, bot, artificial agent, etc is basically software program driven by artificial intelligence which serves the purpose of making a conversation with the user by texts or by speech.

Example:

- ❖ Siri
- ❖ Alexa



These chatbots are inclined towards performing a specific task for the user. Chatbots often perform tasks like making a transaction, booking a hotel, form submissions, etc. The possibilities with a chatbot are endless with the technological advancements in the domain of artificial intelligence.

PROBLEM STATEMENT:

The challenge is to create a Chatbot in Python that provides exceptional customer service, answering user queries on a website or application.

Objectives:

- ❖ Deliver high quality support to the users

- ❖ Ensuring a positive user experience
- ❖ Customer satisfaction

PROBLEM DEFINITION:

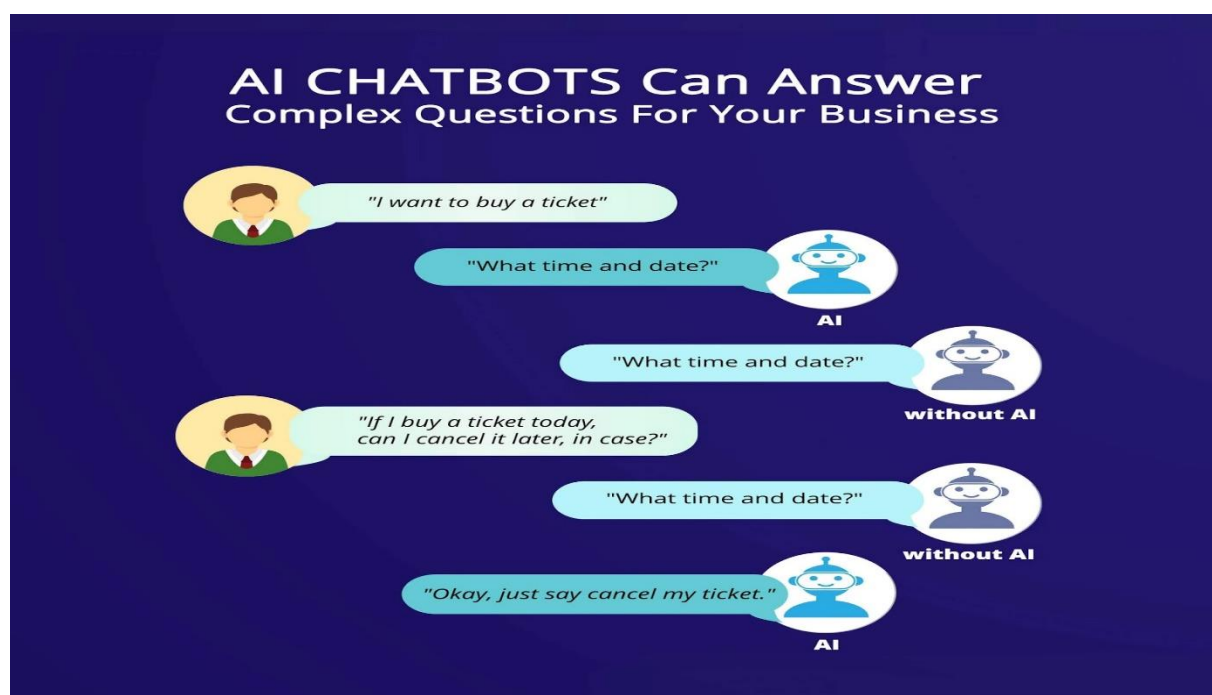
The problem is to build an AI-powered prediction system that uses machine learning algorithms to analyze data and predict the likelihood of an individual developing activities.

The system aims to provide early risk assessment and personalized preventive measures, allowing individuals to take proactive actions to manage their problems.

DESIGN THINKING:

1.Functionality

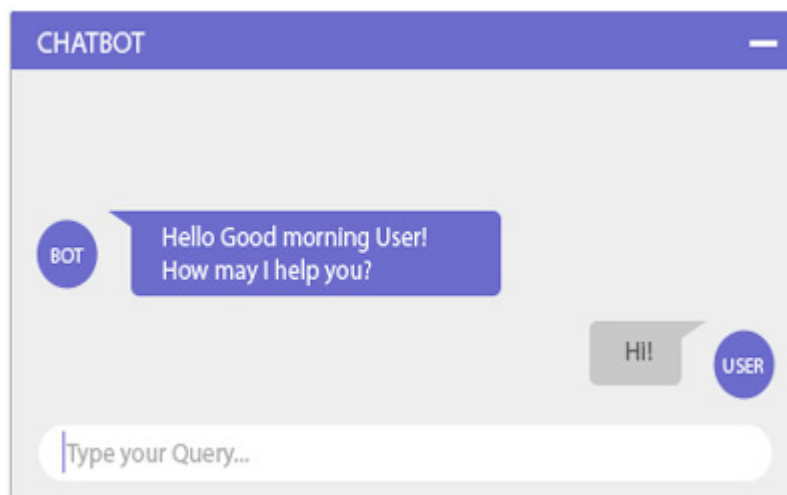
A Chatbot, however, can answer questions 24 hours a day seven days a week. It can provide a guidance, supplement support during peak periods, or offload tedious repetitive questions so human agents can focus on more complex issues.



Chatbots are bots designed to answer common questions people have about a product or service. Companies employ these chatbots for services like customer support, to deliver information, etc.

2.User Interface

Chatbot interface design refers to the form, while chatbot user experience is based on subjective impressions of end-users. If the UI is confusing or difficult to use, users will not be able to communicate with the chatbot effectively. The UI determines how users feel when they are using the chatbot.



3.NLP

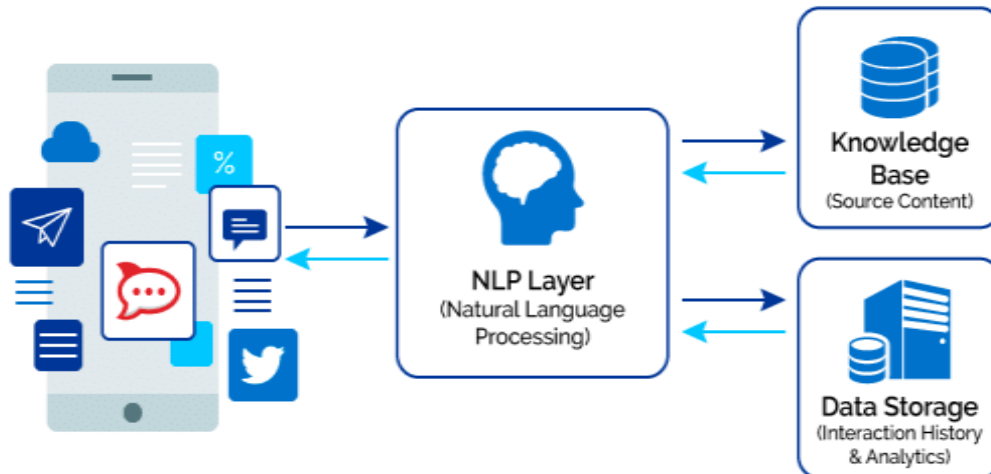
NLP-Natural Language Processing

An natural language processing chatbot is a software program that can understand and respond to human speech. Bots powered by NLP allow people to communicate with computers in a way that feels natural and human-like — mimicking person-to-person conversations.

COMPONENTS OF NLP

- ❖ A Convincing UX
- ❖ A Learning curve
- ❖ Understanding multiple languages

- ❖ Administrative analytics
- ❖ Robotic process automation
- ❖ Voice-enabled chatbots



4.Responses

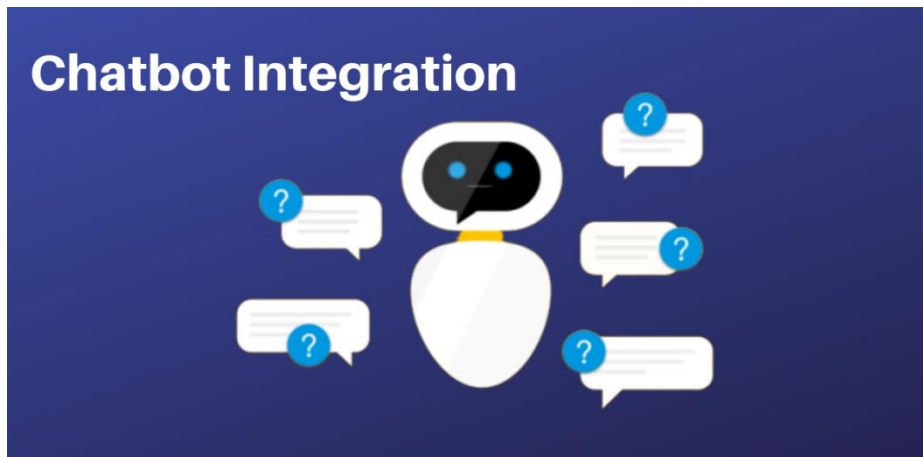
A chatbot response is a message your bot sends to the user. In ChatBot, you can choose from 6 types of bot responses:

- Text
- Random Text
- Image
- Gallery
- Button
- Quick Reply

5.INTEGRATION

An API is a computing interface that allows information sharing between different software components. These interfaces define what kind of requests can be made, how they can be made, which data formats can be used, etc.

In this case, when integrated with chatbots, they allow the expansion of the system functionalities.



EXAMPLE:

Imagine you are talking to the bot and need access to your Client Area. With an integrated API, you can access it directly in the bot without opening another web page.

6.TESTING AND IMPROVEMENT

There comes the need of performance testing. It is not just to measure whether chatbot can sustain its output among a large number of concurrent users, but it's also important to measure response time for an individual user.

Chatbot is primarily used to smoothen user experience, by running navigation tests, you can detect if your chatbot fails to respond to specific requests.

If so, you can set up your chatbot to trigger and answer particular user questions at any step

Elements of chatbot testing

- ① Onboarding
- ② Conversation flow
- ③ Navigation
- ④ Response time and rate
- ⑤ Error management
- ⑥ Channels and integrations
- ⑦ Visuals and links

CONCLUSION:

Chatbots are a type of digital assistant designed to improve business efficiency by automating routine support tasks.

Chatbots have become fixtures in the online retail space to streamline

- Customer support,
- In Industries
- Finance
- Healthcare
- Insurance.